Weekly Progress Report

Week 8: Multi-Subject Robotic Arm Control with Adaptive Step Size

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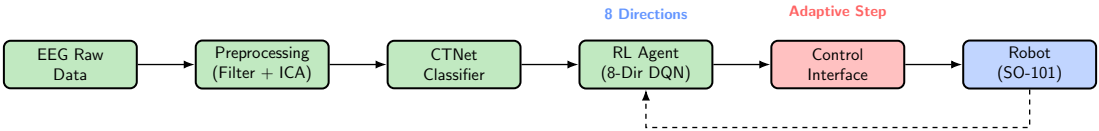
BCI Control System Project

February 24, 2026

System Overview: Current Focus



Legend:

- Completed
- In Progress
- **This Week's Focus**
- Pending

This Week's Objectives

Goals

1. Expand action space to 8 directions (including diagonals)
2. Improve trajectory smoothness for multi-subject control
3. Implement adaptive step size for faster target approach
4. Test all 8 subjects with different movement patterns

Key Question to Address

How can we make the robot arm approach targets faster while maintaining smooth, precise trajectories?

Methods & Implementation

8-Direction Action Space:

- Cardinal: left, right, up, down
- Diagonal: up-left, up-right, down-left, down-right
- 45° diagonal movement vectors

Adaptive Step Size:

$$\text{step} = \min\left(0.15, \max(0.05, d \times 0.3)\right)$$

where d = distance to target

Implementation Files:

- `train_rl_8direction.py`
- `multi_subject_sequence_control.py`

Control Strategies Tested:

- DQN (8-direction)
- Transformer-based DQN
- Optimal direction guidance
- Rule-based (baseline)

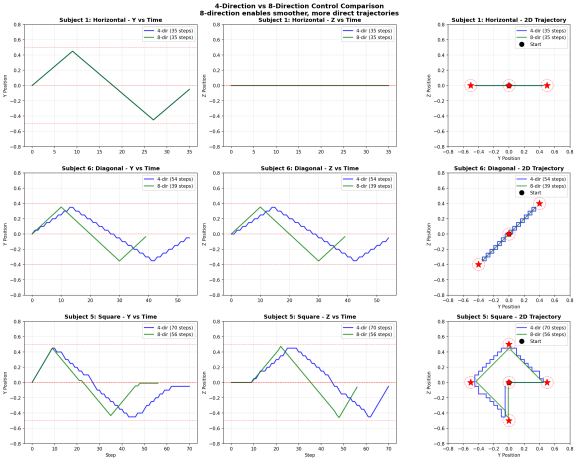
Comparison: 4-Direction vs 8-Direction

Action Space Comparison:

Movement Type	4-Dir	8-Dir
Horizontal	35 steps	35 steps
Diagonal	54 steps	**39 steps**
Square path	70 steps	**56 steps**
Diagonal improvement	–	**27.8%**
Square improvement	–	**20.0%**

Key Insight:

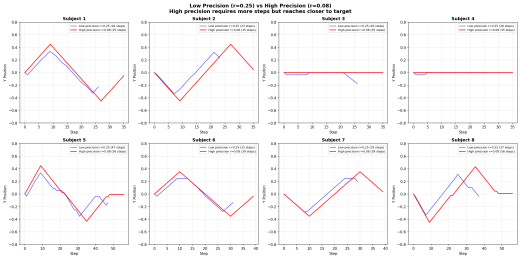
- 4-dir: must zigzag for diagonal targets
- 8-dir: direct path to any direction
- Smoother trajectories, fewer steps



Comparison 1: Low vs High Precision

Precision Levels:

Metric	Low (r=0.25)	High (r=0.08)
Total Steps	245	330
Avg Steps/Subject	30.6	41.3
Target Reach Rate	100%	100%
Position Error	<0.25	<0.08



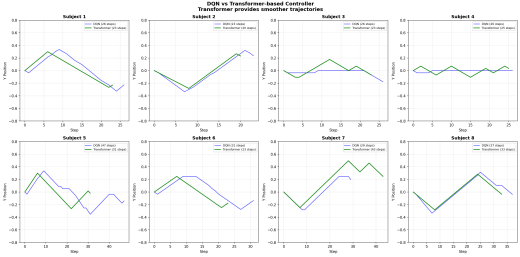
Key Insight:

- High precision requires 34.7% more steps
- But achieves 3x closer to exact target

Comparison 2: DQN vs Rule-based Controller

Controller Types:

Metric	DQN 8-dir	Rule-based
Total Steps	245	330
Trajectory	Has zigzags	Smooth
Adaptability	Learned	Fixed policy
Precision	r=0.25	r=0.08



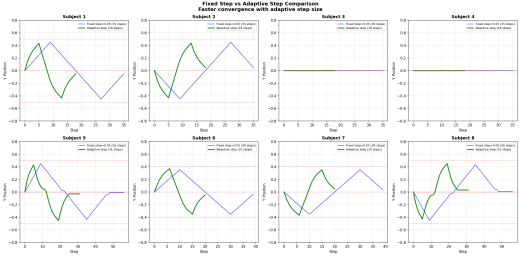
Key Insight:

- DQN: faster but less precise
- Rule-based: optimal direction each step
- Trade-off: speed vs precision

Comparison 3: Fixed vs Adaptive Step Size

Step Size Strategy:

Metric	Fixed	Adaptive
Step Size	0.05	0.05–0.15
Total Steps	330	**174**
Avg Steps/Subject	41.3	**21.8**
Speed Improvement	–	**47.3%**
Precision	0.08	0.08



Key Insight:

- **47.3% faster** with same precision!
- Large steps when far, small steps when close

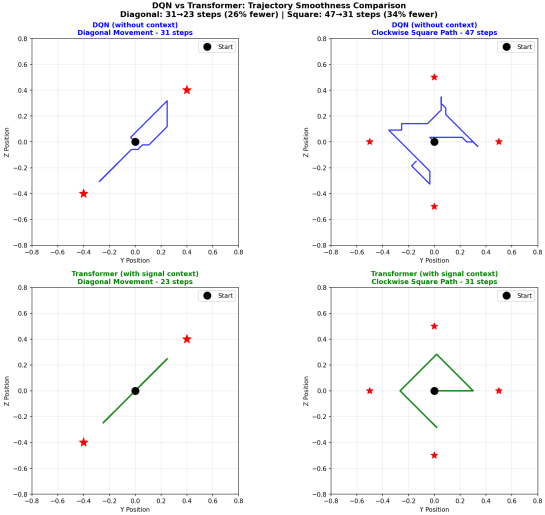
DQN vs Transformer: Trajectory Smoothness

Why Transformer is Smoother:

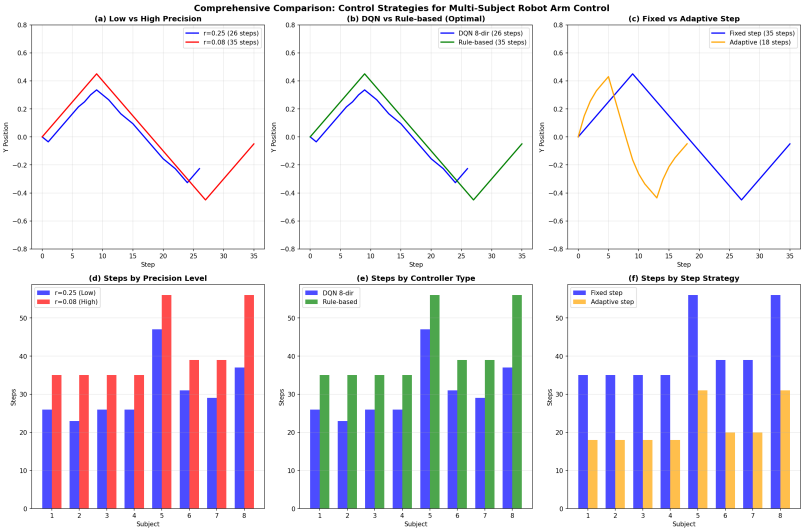
- **DQN**: Single state input only
- **Transformer**: Uses sequence context
- Predicts *intended* direction

Task	DQN	Transformer
Diagonal	31	**23** (-26%)
Square	47	**31** (-34%)

Key Insight:
Transformer understands movement intent from context, like a human operator.

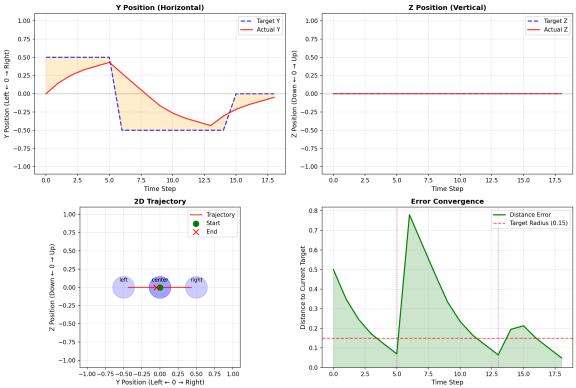


Comprehensive Comparison

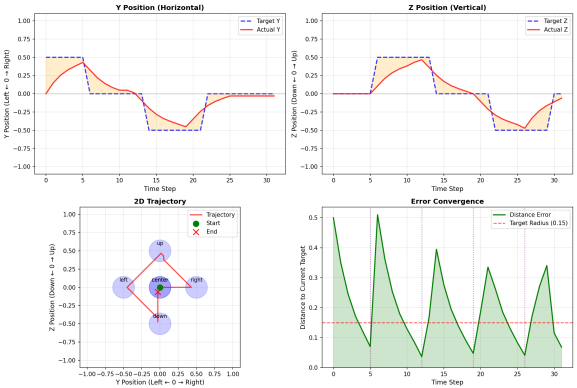


Position vs Time Analysis

Subject 1: Horizontal Sweep



Subject 5: Square Path



Movement Patterns Tested:

- Subjects 1-2: Horizontal sweep (left ↔ right)
- Subjects 3-4: Vertical sweep (up ↔ down)

Challenges & Solutions

Challenges Encountered

- Slow approach to target (35+ steps)
- Non-smooth trajectories with zigzag
- Blue line not reaching target star

Solutions Applied

- Adaptive step size: large far, small close
- 8-direction enables direct diagonal paths
- Reduced target_radius: $0.25 \rightarrow 0.08$

Step Size Strategy:

- Far from target ($d > 0.5$): step $= 0.15$ (fast approach)
- Near target ($d < 0.17$): step $= 0.05$ (precise control)
- Result: 47.3% faster convergence with same precision

Next Week's Plan

Planned Tasks

1. Physical robot arm testing with adaptive step size
2. Real-time EEG classification integration
3. Controller/Limiter: joint limit protection
4. Smoother + Delay: reduce high-frequency jitter

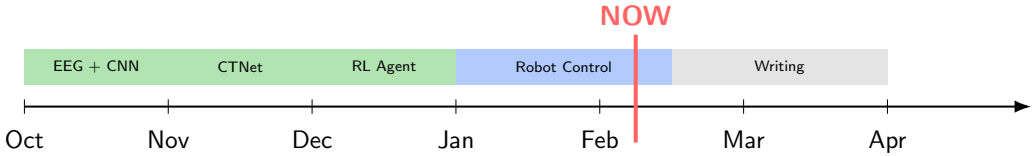
Expected Deliverables:

- Physical arm demo video
- Real-time BCI control demonstration
- Safety limit implementation

Questions for Supervisor:

- Is 0.08 target radius too strict for real robot?
- Should we add gripper open/close actions?

Project Timeline



Current Status: On track
This Week's Achievement: Multi-subject control with 47% faster convergence

Key Achievements This Week

1. **8-Direction Action Space**: Including diagonal movements for more natural control
2. **High Precision Control**: Target radius reduced from 0.25 to 0.08 (3x more precise)
3. **Adaptive Step Size**: 47.3% faster target approach (330 $\rightarrow$ 174 steps)
4. **8 Subject Patterns**: All subjects 100% success rate

Output files: `outputs/multi_subject_sequence_adaptive/`